



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

Deep Learning (Unit-01)



Dr. Zulfikar Ali Ansari

Assistant Professor, Department of AIML



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Course Name: Deep Learning
Course Code: TE7753
Faculty: Engineering
Course Credit: 3
Course Level: 2
Sub-Committee (Specialization): CSE and IT

Learning Objectives:

1. Discuss various Deep Learning concepts, principles, algorithms and concepts
2. Describe all concepts related to NN which will be useful for DL applications
3. Summarize various DL models and deep NN
4. Give examples of various optimization algorithms
5. Discuss and extend CNN and RNN concepts

Books

Recommended:

Book	Author	Publisher
1. Introduction to Deep Learning	Eugene Charniak	MIT Press
2. Deep Learning with Keras	Antonio Gulli, Sujit Pal	Packt.com

Course Outline:

Sr. No.	Topic	Actual Teaching Hours	Contact Hours Equivalence
1	Introduction to Deep Learning - NN, ANN, Why Deep Learning Brief History of AI, Deep Learning: Successes Last Decade, reasons to go Deep, applications of Deep Learning, Challenges of Deep Learning	10	10
2	Advances in NN and DL- Biological Neuron Vs Perceptron, Shallow Neural Network, Training a Perceptron, Linear Classification, Backpropagation, Role of Activation Functions and Backpropagation, Activation Function, Backprop Illustration, Optimization, Regularization, Dropout layer, Classification Kaggle Dataset, FFNN	8	8
3	Deep Neural Network: Why and Applications, Designing a Deep Neural Network, How to Choose Your Loss Function, Tools for Deep Learning Models, Keras and its Elements, Tensorflow and Its Ecosystem, Build a Deep Learning Model Using Tensorflow, TFlearn	10	10
4	Deep Neural Net optimization, tuning, interpretability, Optimization Algorithms, SGD, Momentum, NAG, Adagrad, Adadelta , RMSprop, Adam, Batch Normalization, Exploding and Vanishing Gradients, Hyperparameter Tuning, Interpretability.	9	9
5	Convolutional Neural Net, Success and History, CNN Network Design and Architecture, Image Type Classification (Kaggle), Deep Convolutional Models , Introduction to Recurrent Neural Networks	8	8
Total		45	45

Evaluation Plan

Department: AIML

Batch: 2024-28

Course name: Deep Learning

Credit: 03

Year / Sem: 2026 – Sem 5

Name of the faculty member: Dr. Shrikrishna Kolhar, Dr. Tanveer Ahmed, Dr. Zulfikar Ali Ansari

Continuous Assessment (CA)

Sr. No.	Component	CO	Max marks	Weightage	Planned Date
1	MCQ Quiz	CO1, CO2	8 (CO1-4, CO2-4)	26.66% (CO1-13.3%, CO2-13.3%)	4 August 2026
2	Unit Test	CO1, CO2, CO3	12 (CO1-2, CO2-3, CO3-7)	40.00% (CO1-6.6%, CO2-10%, CO3-23.3%)	7 – 12 September 2026
3	Project Report	CO3, CO4, CO5	10 (CO3-2, CO4-4, CO5-4)	33.33% (CO3-6.6%, CO4-13.3%, CO5-13.3%)	19 October 2026



Signature of Course In-charge
Dr. Shrikrishna Kolhar



Signature of HoD

Evaluation Rubrics

Project Report

Criteria	COs	Max Marks	Excellent (2)	Good (1.5)	Average (1)	Poor (0.5)
Problem Definition and Background (Introduction, Literature review)	CO1	2	Clearly defined problem with strong justification and relevant literature	Clear problem with limited background	Basic problem statement	Unclear / irrelevant
Implementation & Model Development	CO2, CO3	2	Correct model implementation with appropriate architecture	Minor issues in implementation	Partial implementation	Incorrect / missing
Optimization & Experimentation	CO4	2	Effective tuning, optimizer use, proper experiments	Limited tuning/experiments	Minimal experimentation	No optimization
Results Analysis & Interpretation	CO5	2	Insightful analysis with comparisons and conclusions	Basic analysis	Superficial interpretation	No meaningful analysis

Course Outcomes

CO1: Explain the fundamental concepts of deep learning, including neural networks, historical evolution, applications, and challenges.

CO2: Implement shallow neural networks using perceptron models, activation functions, and backpropagation techniques.

CO3: Construct deep neural network models using frameworks such as TensorFlow and Keras for solving real-world problems.

CO4: Optimize deep learning models by applying advanced optimization algorithms, regularization methods, and hyperparameter tuning strategies.

CO5: Analyse convolutional and recurrent neural network architectures for image classification and sequence modelling tasks.

Session Plan		
Name of the department – AIML		
Name of the course- Deep Learning	Credit - 03	
Semester – 5	Batch – 2024-28	
Name of the faculty- Dr. Shrikrishna Kolhar, Dr. Tanveer Ahmed, Dr. Zulfikar Ali-Ansari		

Lect : No	Unit No.	Points to cover	Methodology	Faculty Conducting	Lecture/ Exp. Learning/ Evaluation	CO
1	1	Introduction: NN, ANN – What is a Neural Network, What is Deep Learning	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
2	1	Brief History of AI – From symbolic AI to modern DL milestones	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
3	1	Deep Learning Successes in the Last Decade – Image, Speech, NLP breakthroughs	Lecture + Case Study	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
4	1	Reasons to go Deep – Hierarchical feature learning, representational power	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
5	1	Applications of Deep Learning – CV, NLP, Healthcare, Autonomous systems	Lecture + Video	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
6	1	Challenges of Deep Learning – Data, compute, overfitting, interpretability	Lecture + Discussion	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
7	1	ANN Architecture – Layers, weights, biases, forward pass	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO1
8	1	ANN Hands-on – Build a simple ANN in Python/Keras	Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia l Learning	CO1
9	1	ANN Hands-on – Build a simple ANN in Python/Keras	Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia l Learning	CO1
10	2	Biological Neuron, McCulloch-Pitts model, Shallow Neural Network – Single hidden layer, forward propagation	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO2
11	2	Training a Perceptron – Perceptron learning rule, convergence	Lecture + Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO2

12	2	Linear Classification & Backpropagation – Chain rule, gradient flow	Lecture + Whiteboard	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO2
13	2	Activation Functions – Sigmoid, ReLU, Tanh, Leaky ReLU, Softmax	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO2
14	2	Backprop Illustration – Worked numerical example, computational graph	Lecture + Whiteboard	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia 1 Learning	CO2
15	2	Optimization, Regularization, Dropout – L1/L2 regularization, Dropout layer	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO2
16	2	Classification on Kaggle Dataset + FFNN Implementation	Lab / Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia 1 Learning	CO2
17		CA1 – MCQ Quiz CO1, CO2 10 Marks Date: 10 August 2026	Written Quiz	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Evaluation	CO1, CO2
18	3	Why Deep Neural Networks? – Universal approximation, depth vs width	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
19	3	Applications of Deep NN – Real-world use cases, benchmarks	Lecture + Case Study	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
20	3	Designing a Deep Neural Network – Architecture choices, depth, width, skip connections	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
21	3	How to Choose Your Loss Function – MSE, Cross-Entropy, Hinge, custom losses	Lecture + Examples	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
22	3	Tools for Deep Learning – Overview of frameworks: TensorFlow, Keras, PyTorch	Lecture + Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
23	3	Keras and its Elements – Sequential API, Functional API, layers, callbacks	Lecture + Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
24	3	TensorFlow and Its Ecosystem	Lecture + PPT	Dr. Shrikrishna Kolhar	Experientia	CO3

25	3	Build a Deep Learning Model Using TensorFlow – Step-by-step hands-on	Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experiential Learning	CO3
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26	3	TFLearn – High-level API on TF, building models with TFLearn	Lecture + Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
27	3	Review Units 1, 2 and 3 recap, Q&A, Unit Test Preparation guidelines	Discussion	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO3
28	2	CA2 – Unit Test CO1, CO2, CO3 12 Marks Date: 7–12 September 2026	Written Test	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Evaluation	CO1, CO2, CO3
29	4	Optimization Algorithms – Overview: SGD, Momentum, Nesterov Accelerated Gradient (NAG)	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4

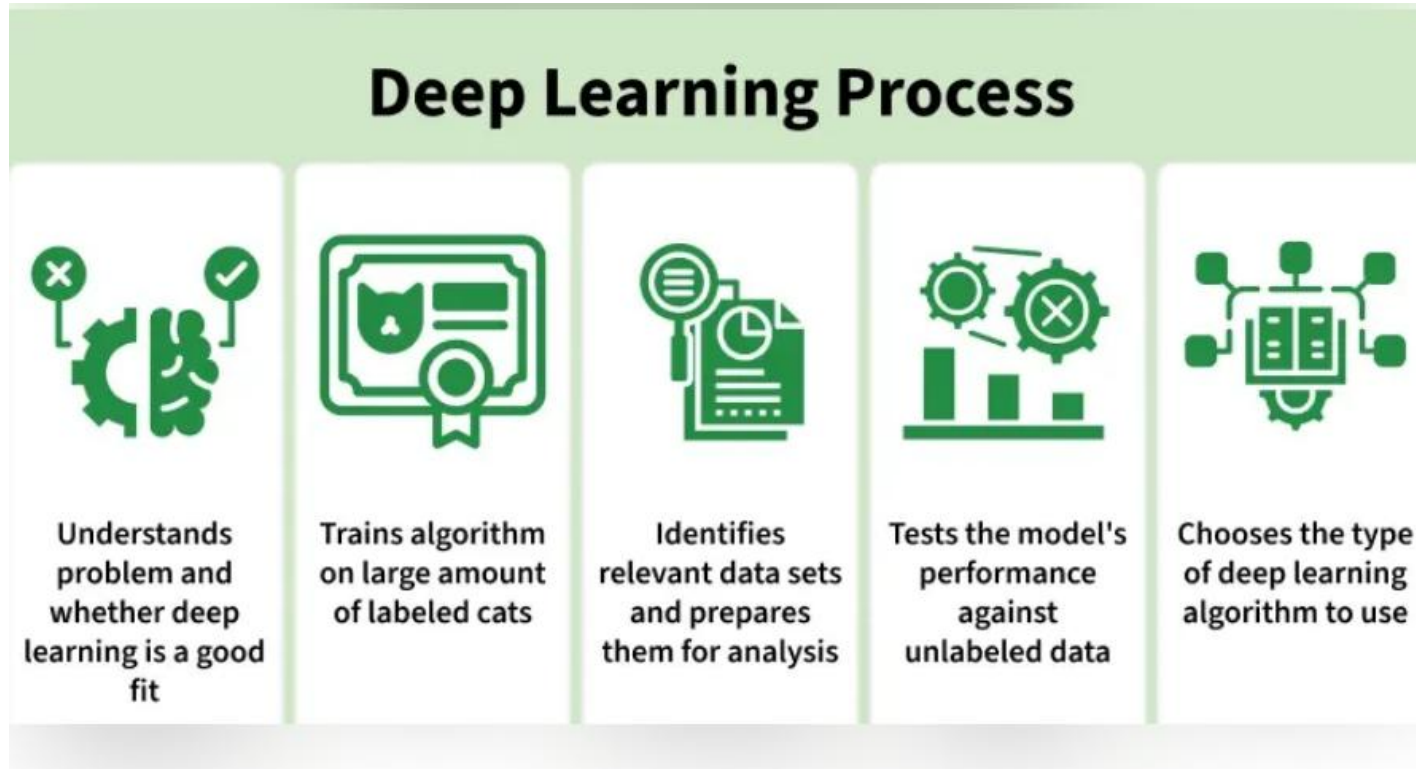
30	4	Adagrad, Adadelata, RMSprop – Adaptive learning rate methods	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
31	4	Adam Optimizer – Combining Momentum and RMSprop, hyperparameters	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
32	4	Batch Normalization – Covariate shift, BN layer, training vs inference	Lecture + Whiteboard	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
33	4	Exploding and Vanishing Gradients – Causes, gradient clipping, weight init strategies	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
34	4	Hyperparameter Tuning – Learning rate, batch size, epochs, grid/random search	Lecture + Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
35	4	Interpretability – LIME, SHAP, saliency maps, Grad-CAM overview	Lecture + Case Study	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO4
36	4	Optimizer Comparison– Comparison of SGD vs Adam vs RMSprop	Coding	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia l Learning	CO4
37	5	CNN: Success and History – LeNet, AlexNet, VGG, ResNet milestones	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experientia l Learning	CO5
38	5	CNN Network Architecture – Convolution, pooling, stride, padding, feature maps	Lecture + Diagram	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5

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39	5	CNN Design Principles – Depth, filter size, receptive field, parameter sharing	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5
40	5	Image Classification on Kaggle – CNN pipeline: preprocess → build → train → evaluate	Coding Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Experiential Learning	CO5
41	5	Deep Convolutional Models – ResNet, Inception, MobileNet: skip connections, depthwise conv	Lecture + Case Study	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5
42	5	Transfer Learning with Pretrained CNN – Fine-tuning, feature extraction	Lecture + Demo	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5
43	5	Introduction to RNN – Sequential data, hidden state, unfolding through time	Lecture + PPT	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5
44	5	RNN Variants – LSTM, GRU – gates, memory cells, vanishing gradient solution	Lecture + Diagram	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Lecture	CO5
45	4	CA3 – Project Report CO1–CO5 8 Marks Date: 10 October 2026	Project Submission	Dr. Shrikrishna Kolhar / Dr. Tanveer Ahmed / Dr. Zulfikar Ali-Ansari	Evaluation	CO4 & CO5

Introduction to Deep Learning

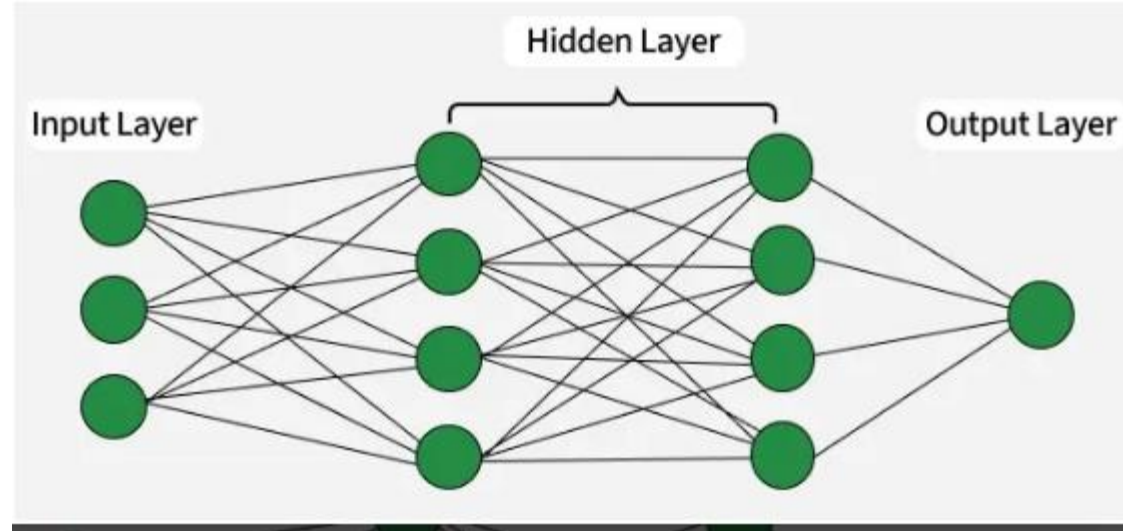
Deep Learning is transforming the way machines understand, learn and interact with complex data. Deep learning mimics neural networks of the human brain, it enables computers to autonomously uncover patterns and make informed decisions from vast amounts of unstructured data.



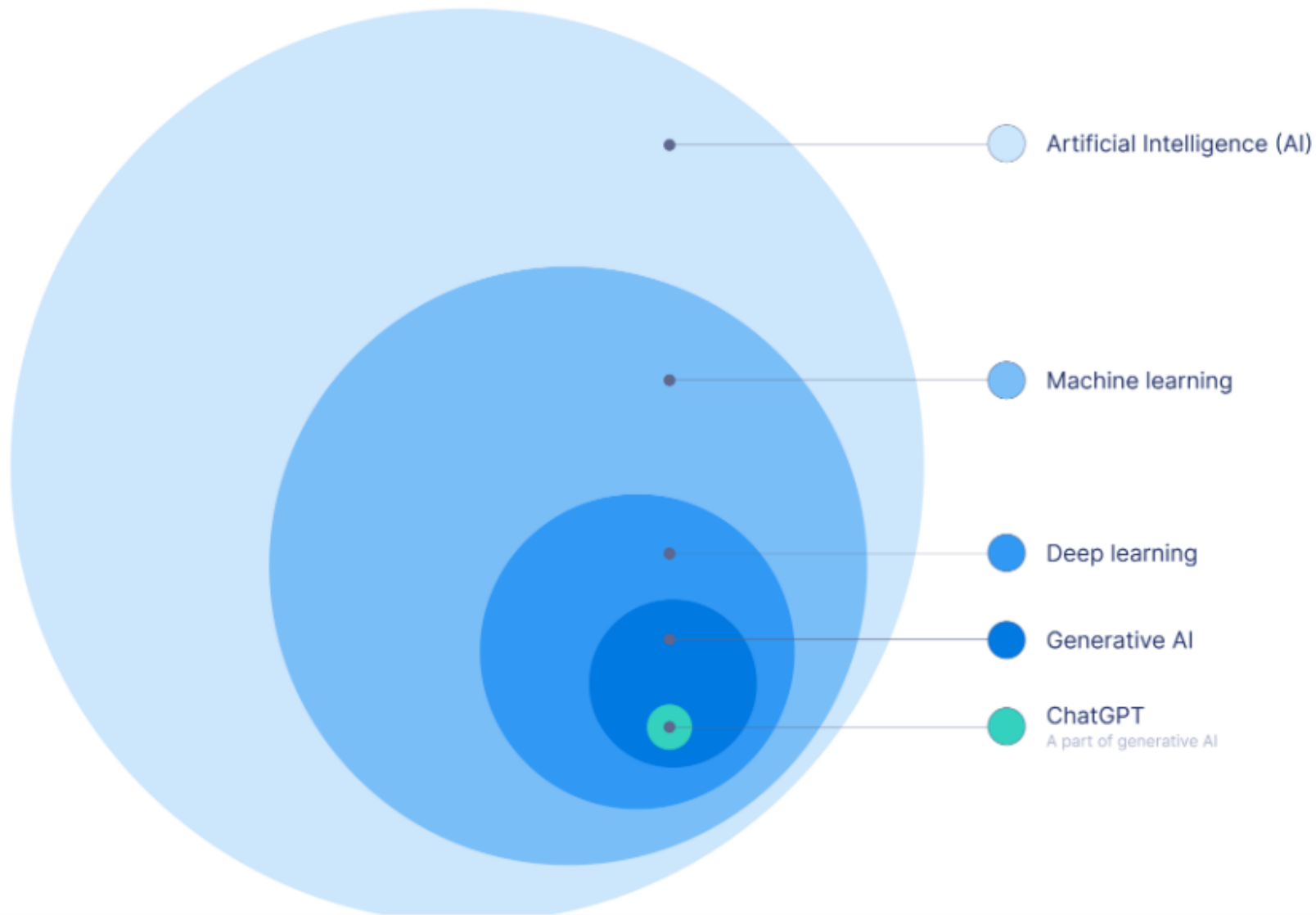
Working of Deep Learning

Neural network consists of layers of interconnected nodes or neurons that collaborate to process input data. In a fully connected deep neural network data flows through multiple layers where each neuron performs nonlinear transformations, allowing the model to learn intricate representations of the data.

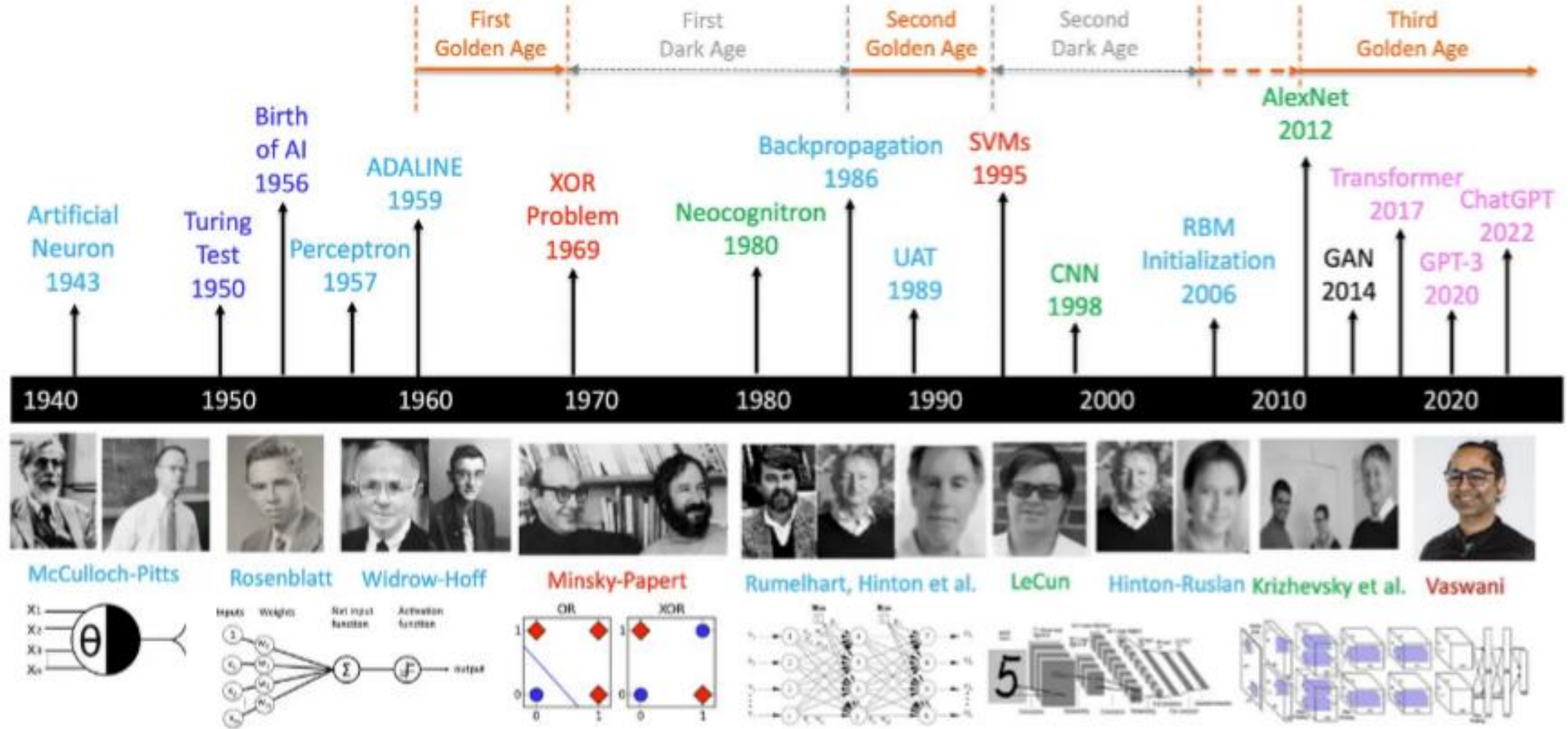
In a deep neural network the input layer receives data which passes through hidden layers that transform the data using nonlinear functions. The final output layer generates the model's prediction.



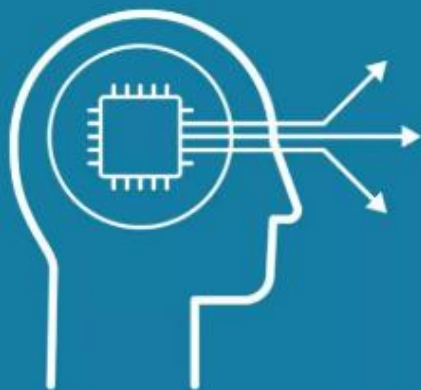
The AI Spectrum: Unveiling Layers of Intelligent Systems



A Brief History of AI with Deep Learning



1950



ARTIFICIAL INTELLIGENCE

ENGINEERING OF MACHINES
THAT MIMIC COGNITIVE FUNCTIONS

1980



MACHINE LEARNING

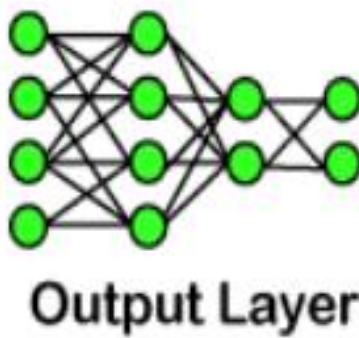
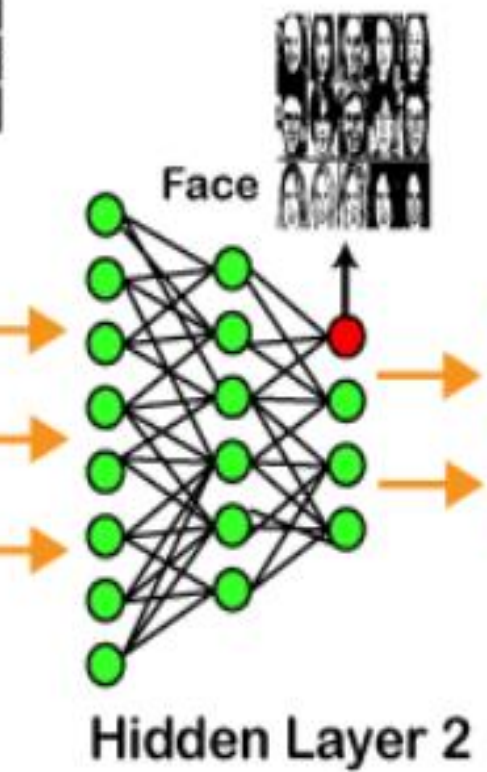
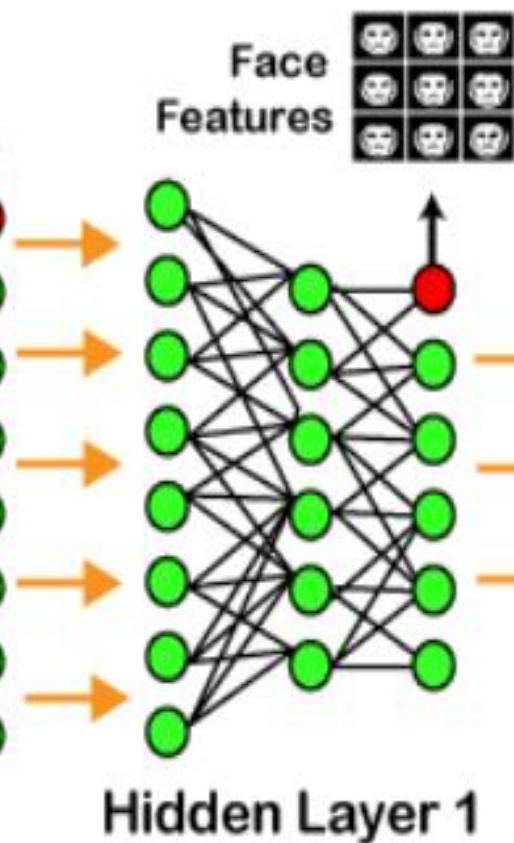
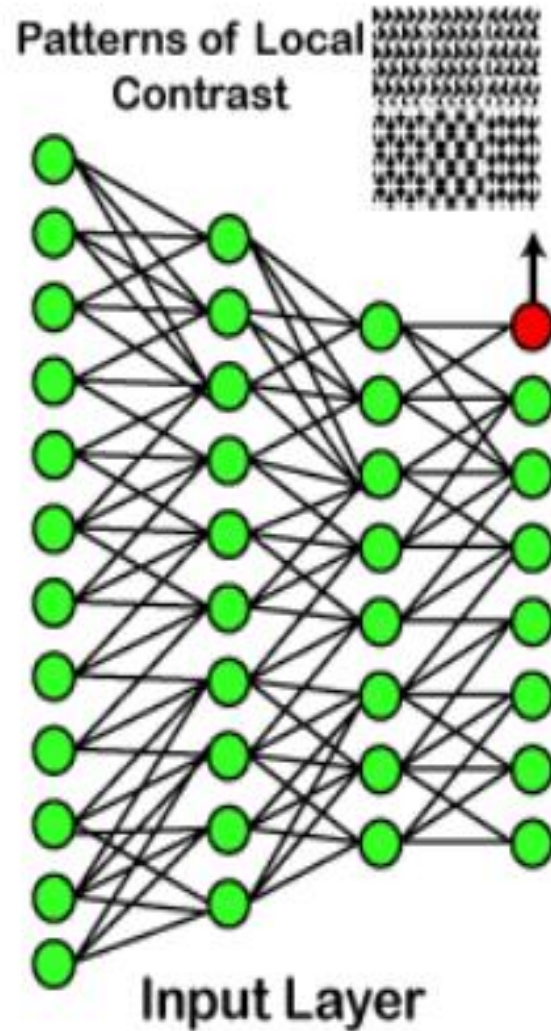
ABILITY TO PERFORM TASKS
WITHOUT EXPLICIT INSTRUCTIONS
AND RELYING ON PATTERNS

2010

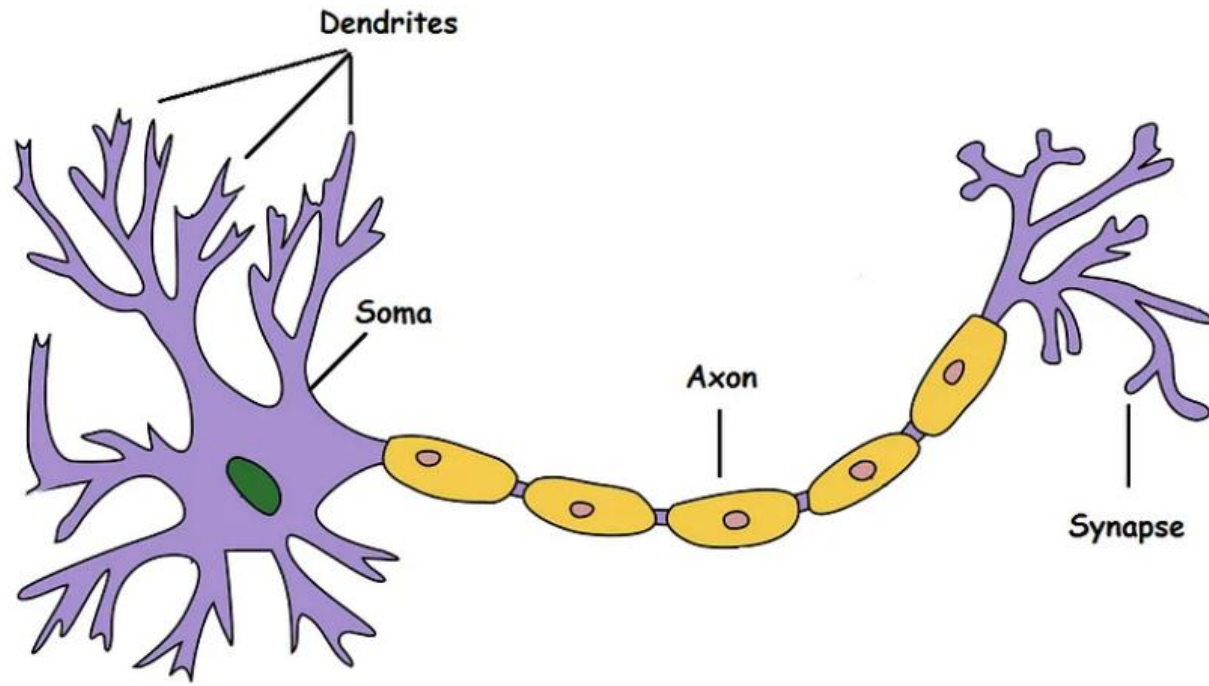


DEEP LEARNING

MACHINE LEARNING BASED
ON ARTIFICIAL NEURAL NETWORKS



Biological inspiration

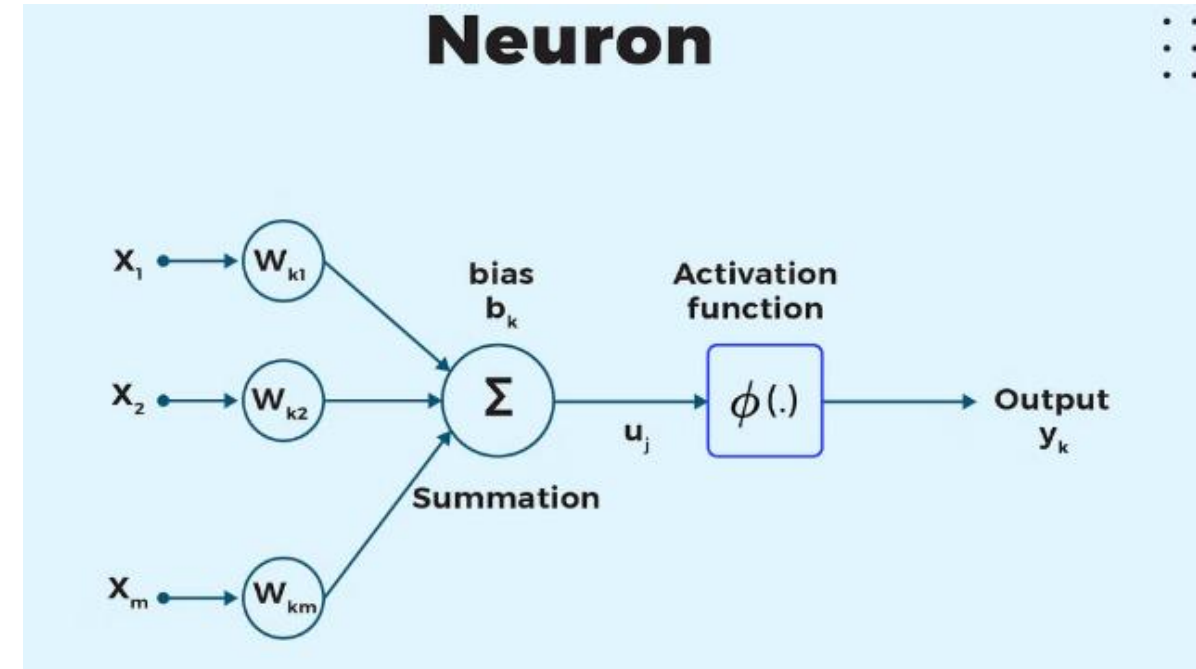


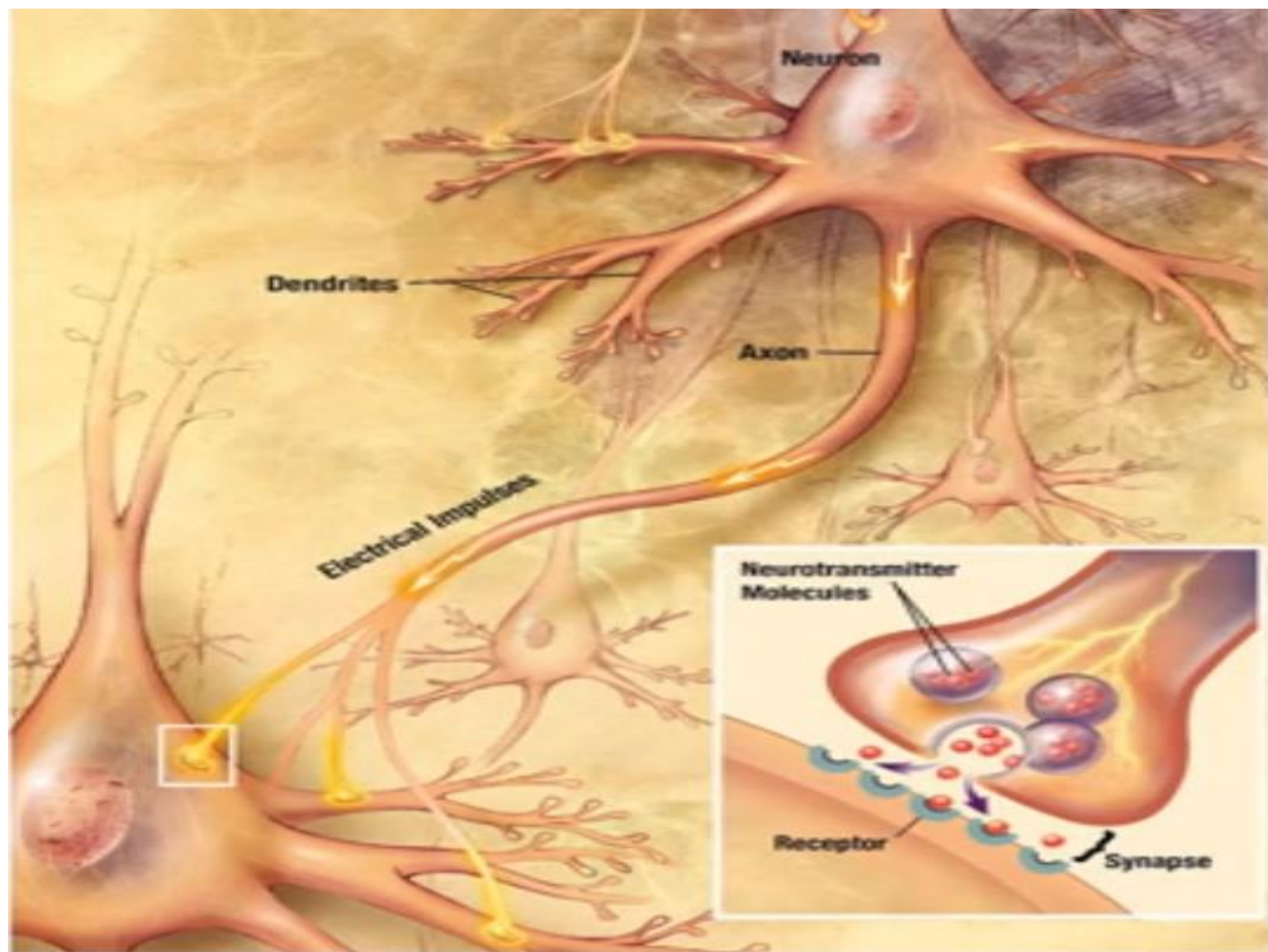
Dendrite: Receives signals from other neurons

Soma: Processes the information

Axon: Transmits the output of this neuron

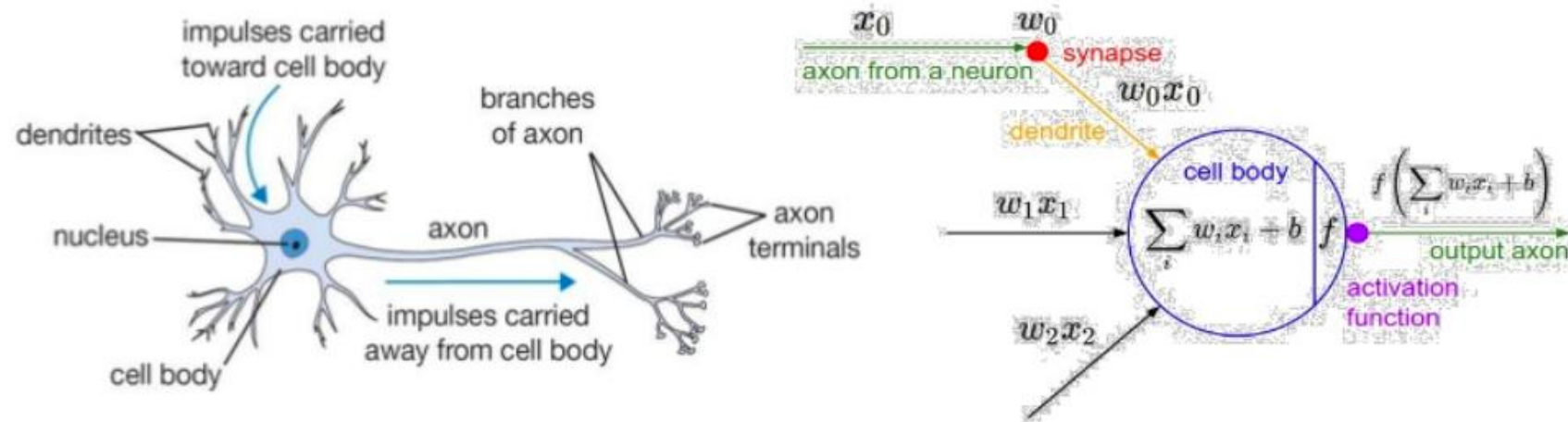
Synapse: Point of connection to other neurons





Deep Learning - Basics

The Neuron



An artificial neuron contains a **nonlinear activation function** and has several incoming and outgoing **weighted connections**.



Neurons are **trained to filter and detect specific features** or patterns (e.g. edge, nose) by receiving weighted input, transforming it with the activation function and passing it to the outgoing connections.

Aspect	Machine Learning	Deep Learning
Basic Idea	Applies statistical algorithms to learn patterns from data	Uses artificial neural networks to learn patterns from data
Data Requirement	Works well with small to medium datasets	Requires a large amount of data
Task Complexity	Better for simple and low-label tasks	Better for complex tasks like image and text processing
Training Time	Takes less time to train	Takes more time to train
Feature Extraction	Features are manually selected and extracted	Features are automatically extracted
Learning Process	Not end-to-end	End-to-end learning
Model Complexity	Less complex	Highly complex
Interpretability	Easy to understand and explain	Hard to interpret (black box)
Hardware Requirement	Can run on CPU, needs less computing power	Needs GPU and high-performance systems
Use Cases	Spam detection, recommendation systems	Image recognition, NLP, speech recognition

Types of neural networks

1.Feedforward neural networks (FNNs): They are the simplest type of ANN, where data flows in one direction from input to output. It is used for basic tasks like classification.

2.Convolutional Neural Networks (CNNs): They are specialized for processing grid-like data, such as images. CNNs use convolutional layers to detect spatial hierarchies, making them ideal for computer vision tasks.

3.Recurrent Neural Networks (RNNs): They are able to process sequential data, such as time series and natural language. RNNs have loops to retain information over time, enabling applications like language modeling and speech recognition. Variants like LSTMs and GRUs address vanishing gradient issues.

4.Generative Adversarial Networks (GANs): This consist of two networks a generator and a discriminator that compete to create realistic data. GANs are widely used for image generation, style transfer and data augmentation.

5.Autoencoders: They are unsupervised networks that learn efficient data encodings. They compress input data into a latent representation and reconstruct it, useful for dimensionality reduction and anomaly detection.

6.Transformer Networks: It has revolutionized NLP with self-attention mechanisms. Transformers excel at tasks like translation, text generation and sentiment analysis, powering models like GPT and BERT.

Applications

1. Computer vision

In computer vision, deep learning models enable machines to identify and understand visual data. Some of the main applications of deep learning in computer vision include:

- Object detection and recognition:** Deep learning models are used to identify and locate objects within images and videos, making it possible for machines to perform tasks such as self-driving cars, surveillance and robotics.
- Image classification:** Deep learning models can be used to classify images into categories such as animals, plants and buildings. This is used in applications such as medical imaging, quality control and image retrieval.
- Image segmentation:** Deep learning models can be used for image segmentation into different regions, making it possible to identify specific features within images.

2. Natural language processing (NLP)

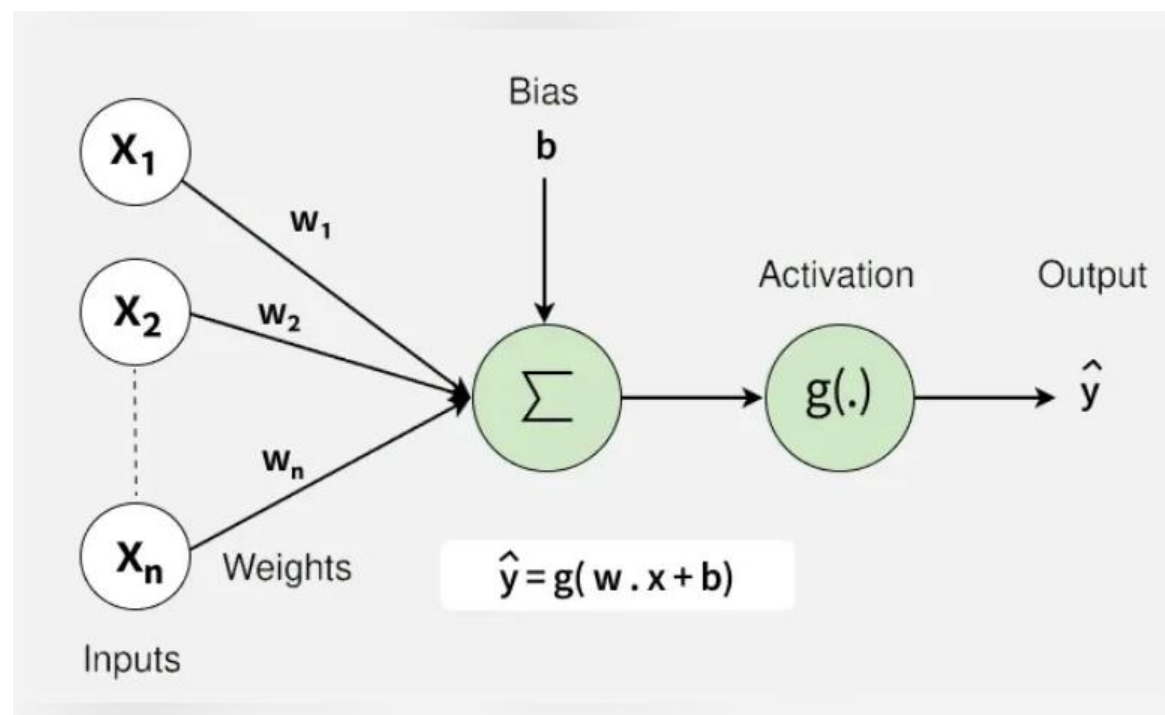
In NLP, deep learning model enable machines to understand and generate human language. Some of the main applications of deep learning in NLP include:

- Automatic Text Generation:** Deep learning model can learn the corpus of text and new text like summaries, essays can be automatically generated using these trained models.
- Language translation:** Deep learning models can translate text from one language to another, making it possible to communicate with people from different linguistic backgrounds.
- Sentiment analysis:** Deep learning models can analyze the sentiment of a piece of text, making it possible to determine whether the text is positive, negative or neutral.
- Speech recognition:** Deep learning models can recognize and transcribe spoken words, making it possible to perform tasks such as speech-to-text conversion, voice search and voice-controlled devices.

Introduction To Neural Networks

A **Neural Network (NN)** is a computational model inspired by the **human brain**. It is designed to recognize patterns, learn from data, and make intelligent decisions.

Neural Networks are the foundation of Deep Learning.



History And Evolution Of Neural Networks

01

1943

First Artificial Neuron Model By McCulloch And Pitts

02

1958

Rosenblatt Develops The Perceptron

03

1970

AI Winter Due To Perceptron Limitations

04

1986

Hinton Introduces Backpropagation

05

2012

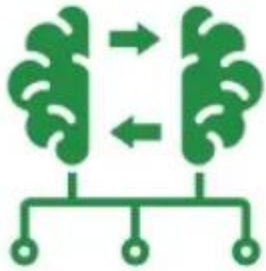
CNN Wins ImageNet Competition, Boosting Deep Learning

06

2020

Transformers Advance AI In NLP And Generative Models

TYPES OF NEURAL NETWORKS



Feedforward
neural networks
(FNNs)



Convolutional
Neural Networks
(CNNs)



Recurrent Neural
Networks (RNNs)



Generative
Adversarial
Networks (GANs)



Autoencoders,
Transformer
Networks

Key Applications of Deep Learning

- 01 Data Science
- 02 Cyber Security
- 03 Autonomous Vehicles
- 04 Social Media Analytics
- 05 Healthcare and Medical Sector
- 06 Computer Vision
- 07 NLP, Speech and Audio

Neural networks are capable of learning and identifying patterns directly from data without pre-defined rules. These networks are built from several key components:

- Neurons:** The basic units that receive inputs, each neuron is governed by a threshold and an activation function.

- Connections:** Links between neurons that carry information, regulated by weights and biases.

- Weights and Biases:** These parameters determine the strength and influence of connections.

- Propagation Functions:** Mechanisms that help process and transfer data across layers of neurons.

- Learning Rule:** The method that adjusts weights and biases over time to improve accuracy.

Learning in neural networks follows a structured, three-stage process:

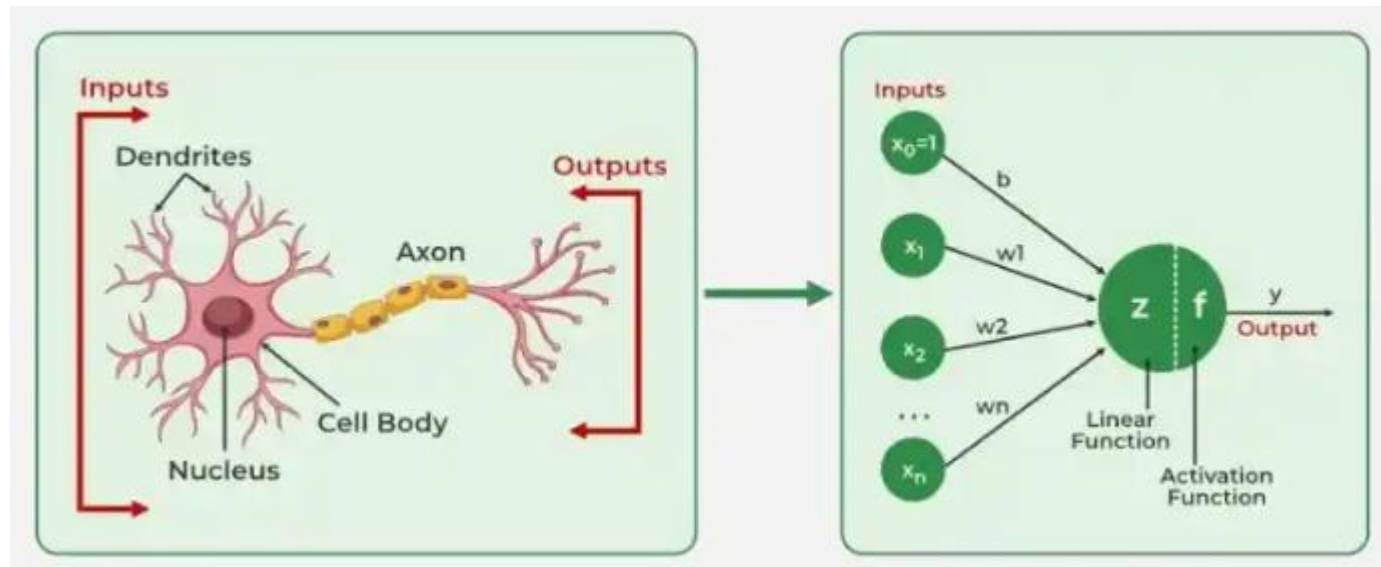
1.Input Computation: Data is fed into the network.

2.Output Generation: Based on the current parameters, the network generates an output.

3.Iterative Refinement: The network refines its output by adjusting weights and biases, gradually improving its performance on diverse tasks.

In an adaptive learning environment:

- The neural network is exposed to a simulated scenario or dataset.
- Parameters such as weights and biases are updated in response to new data or conditions.
- With each adjustment, the network's response evolves allowing it to adapt effectively to different tasks or environments.



Importance of Neural Networks

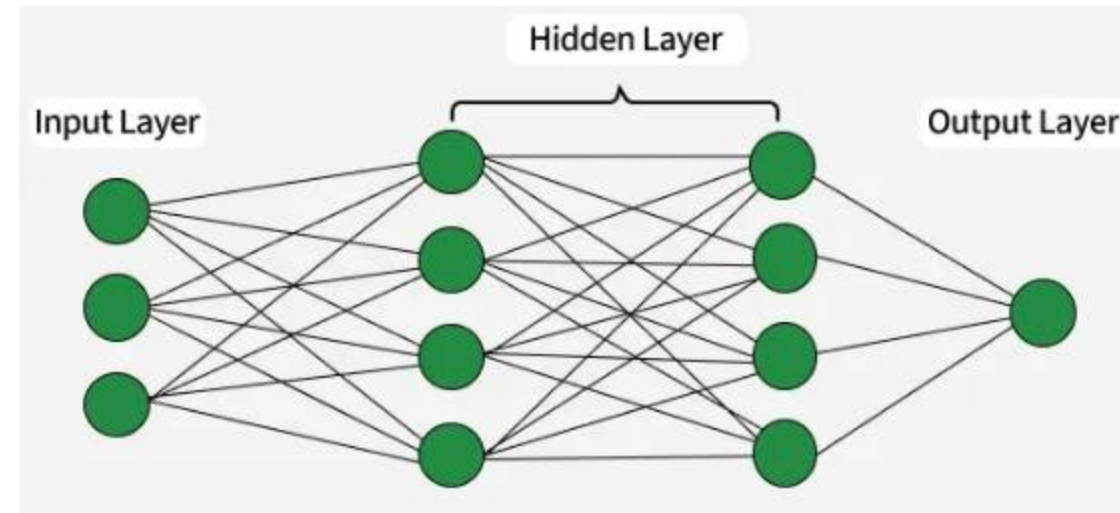
- **Identify Complex Patterns:** Recognize intricate structures and relationships in data; adapt to dynamic and changing environments.
- **Learn from Data:** Handle vast datasets efficiently; improve performance with experience and retraining.
- **Drive Key Technologies:** Power natural language processing (NLP); enable self-driving vehicles; support automated decision-making systems.
- **Boost Efficiency:** Streamline workflows and processes; enhance productivity across industries.
- **Backbone of AI:** Serve as the core driver of artificial intelligence progress; continue shaping the future of technology and innovation.

Layers in Neural Network Architecture

1.Input Layer: This is where the network receives its input data. Each input neuron in the layer corresponds to a feature in the input data.

2.Hidden Layers: These layers perform most of the computational heavy lifting. A neural network can have one or multiple hidden layers. Each layer consists of units (neurons) that transform the inputs into something that the output layer can use.

3.Output Layer: The final layer produces the output of the model. The format of these outputs varies depending on the specific task like classification, regression.



Working of Neural Networks

1. Forward Propagation

When data is input into the network, it passes through the network in the forward direction, from the input layer through the hidden layers to the output layer. This process is known as forward propagation. Here's what happens during this phase:

1. Linear Transformation: Each neuron in a layer receives inputs which are multiplied by the weights associated with the connections. These products are summed together and a bias is added to the sum. This can be represented mathematically as:

$$z = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

where

- w represents the weights
- x represents the inputs
- b is the bias

2. Activation: The result of the linear transformation (denoted as z) is then passed through an activation function. The activation function is crucial because it introduces non-linearity into the system, enabling the network to learn more complex patterns. Popular activation functions include ReLU, sigmoid and tanh.

2. Backpropagation

After forward propagation, the network evaluates its performance using a loss function which measures the difference between the actual output and the predicted output. The goal of training is to minimize this loss. This is where backpropagation comes into play:

- Loss Calculation:** The network calculates the loss which provides a measure of error in the predictions. The loss function could vary; common choices are **mean squared error** for regression tasks or **cross-entropy loss** for classification.

- Gradient Calculation:** The network computes the gradients of the loss function with respect to each weight and bias in the network. This involves applying the chain rule of calculus to find out how much each part of the output error can be attributed to each weight and bias.

- Weight Update:** Once the gradients are calculated, the weights and biases are updated using an optimization algorithm like stochastic gradient descent (SGD). The weights are adjusted in the opposite direction of the gradient to minimize the loss. The size of the step taken in each update is determined by the learning rate.

3. Iteration

This process of forward propagation, loss calculation, backpropagation and weight update is repeated for many iterations over the dataset. Over time, this iterative process reduces the loss and the network's predictions become more accurate.

Through these steps, neural networks can adapt their parameters to better approximate the relationships in the data, thereby improving their performance on tasks such as classification, regression or any other predictive modeling.

Artificial Neural Network (ANN)

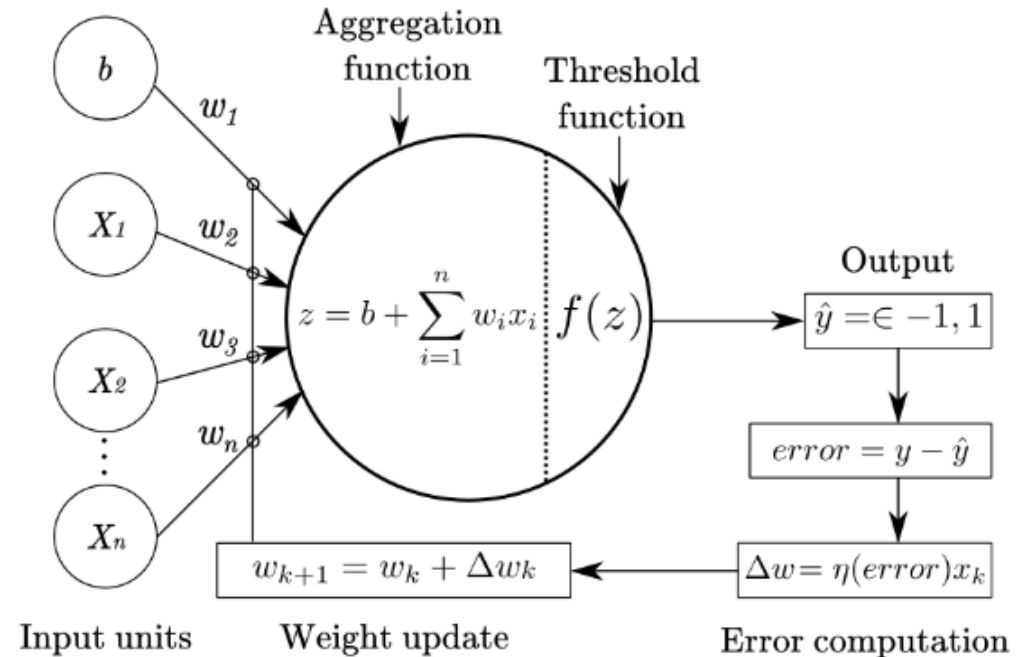
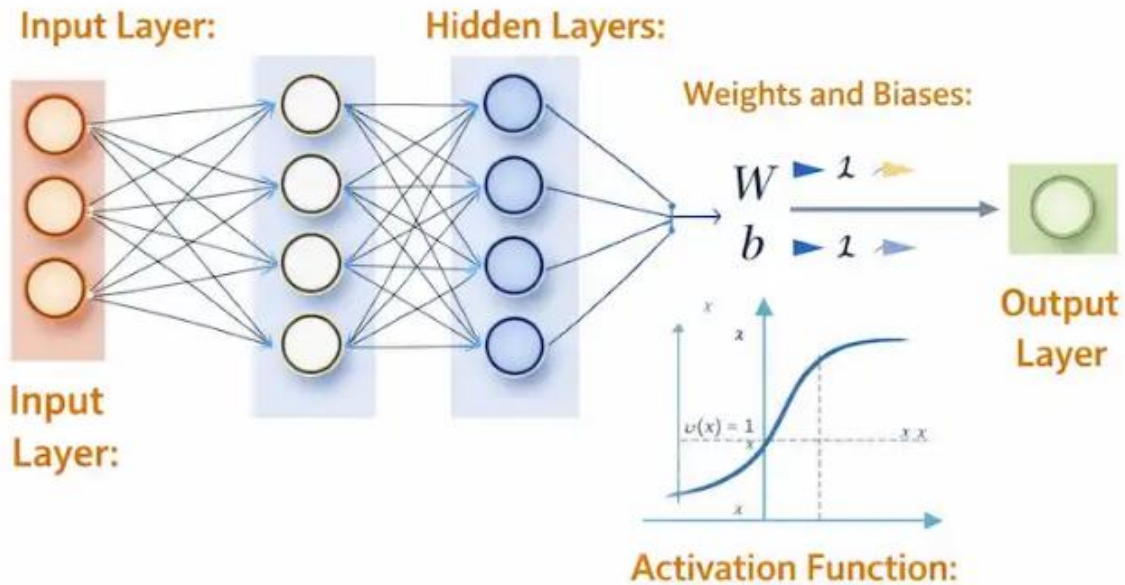
ANN stands for **Artificial Neural Network**.

It is a computer-based model inspired by the human brain's biological neural network.

The word “**Artificial**” is used because:

- biological neurons exist in the human brain
- artificial neurons are mathematical/computational models created in computers

Key Components of ANN Architecture



Key Components of an ANN

- Input Layer:** This is where the network receives information. For example, in an image recognition task, the input could be an image.
- Hidden Layers:** These layers process the data received from the input layer. The more hidden layers there are, the more complex patterns the network can learn and understand. Each hidden layer transforms the data into more abstract information.
- Output Layer:** This is where the final decision or prediction is made. For example, after processing an image, the output layer might decide whether it's a cat or a dog.

Working of Artificial Neural Networks

ANNs work by learning patterns in data through a process called training. During training, the network adjusts itself to improve its accuracy by comparing its predictions with the actual results.

Lets see how the learning process works:

- Input Layer:** Data such as an image, text or number is fed into the network through the input layer.

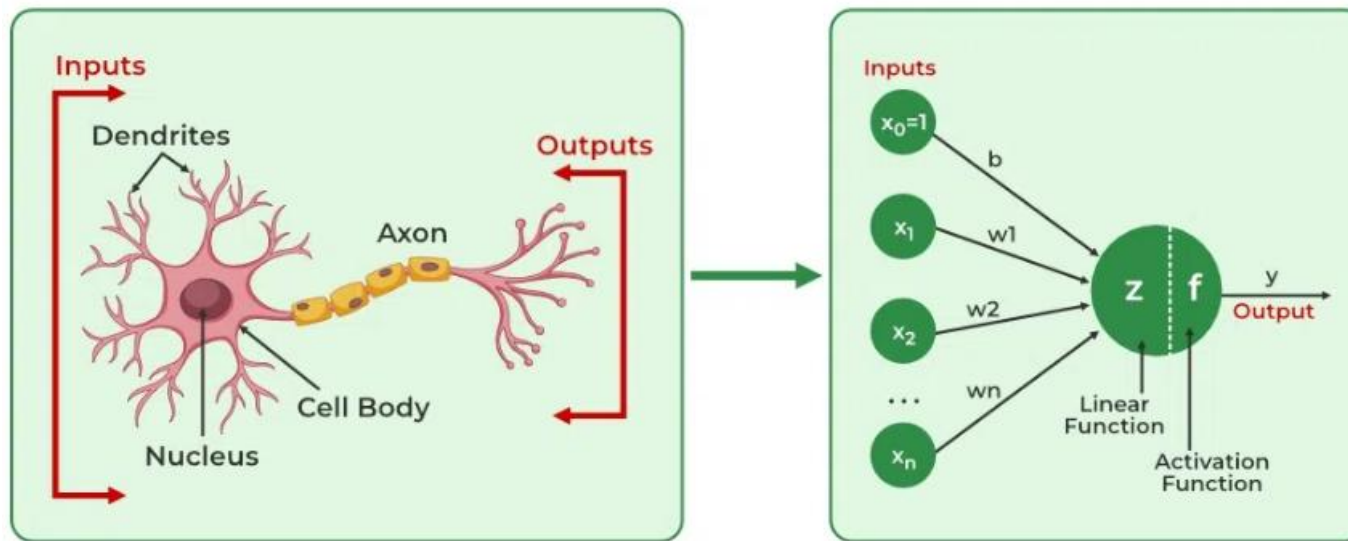
- Hidden Layers:** Each neuron in the hidden layers performs some calculation on the input, passing the result to the next layer. The data is transformed and abstracted at each layer.

- Output Layer:** After passing through all the layers, the network gives its final prediction like classifying an image as a cat or a dog.

The process of backpropagation is used to adjust the weights between neurons. When the network makes a mistake, the weights are updated to reduce the error and improve the next prediction.

How do Artificial Neural Networks learn?

- Artificial Neural Networks (ANNs) learn by training on a set of data. For example, to teach an ANN to recognize a cat, we show it thousands of images of cats. The network processes these images and learns to identify the features that define a cat.
- Once the network has been trained, we test it by providing new images to see if it can correctly identify cats. The network's prediction is then compared to the actual label (whether it's a cat or not). If it makes an incorrect prediction, the network adjusts by fine-tuning the weights of the connections between neurons using a process called backpropagation. This involves correcting the weights based on the difference between the predicted and actual result.
- This process repeats until the network can accurately recognize a cat in an image with minimal error. Essentially, through constant training and feedback, the network becomes better at identifying patterns and making predictions.



Aspect	Biological Neurons	Artificial Neurons
Structure	Dendrites: Receive signals from other neurons.	Input Nodes: Receive data and pass it on to the next layer.
	Cell Body (Soma): Processes the signals.	Hidden Layer Nodes: Process and transform the data.
	Axon: Transmits processed signals to other neurons.	Output Nodes: Produce the final result after processing.
Connections	Synapses: Links between neurons that transmit signals.	Weights: Connections between neurons that control the influence of one neuron on another.
Learning Mechanism	Synaptic Plasticity: Changes in synaptic strength based on activity over time.	Backpropagation: Adjusts the weights based on errors in predictions to improve future performance.
Activation	Activation: Neurons fire when signals are strong enough to reach a threshold.	Activation Function: Maps input to output, deciding if the neuron should fire based on the processed data.

Common Activation Functions in ANNs

Activation functions are important in neural networks because they introduce non-linearity and helps the network to learn complex patterns. Lets see some common activation functions used in ANNs:

1.**Sigmoid Function**: Outputs values between 0 and 1. It is used in binary classification tasks like deciding if an image is a cat or not.

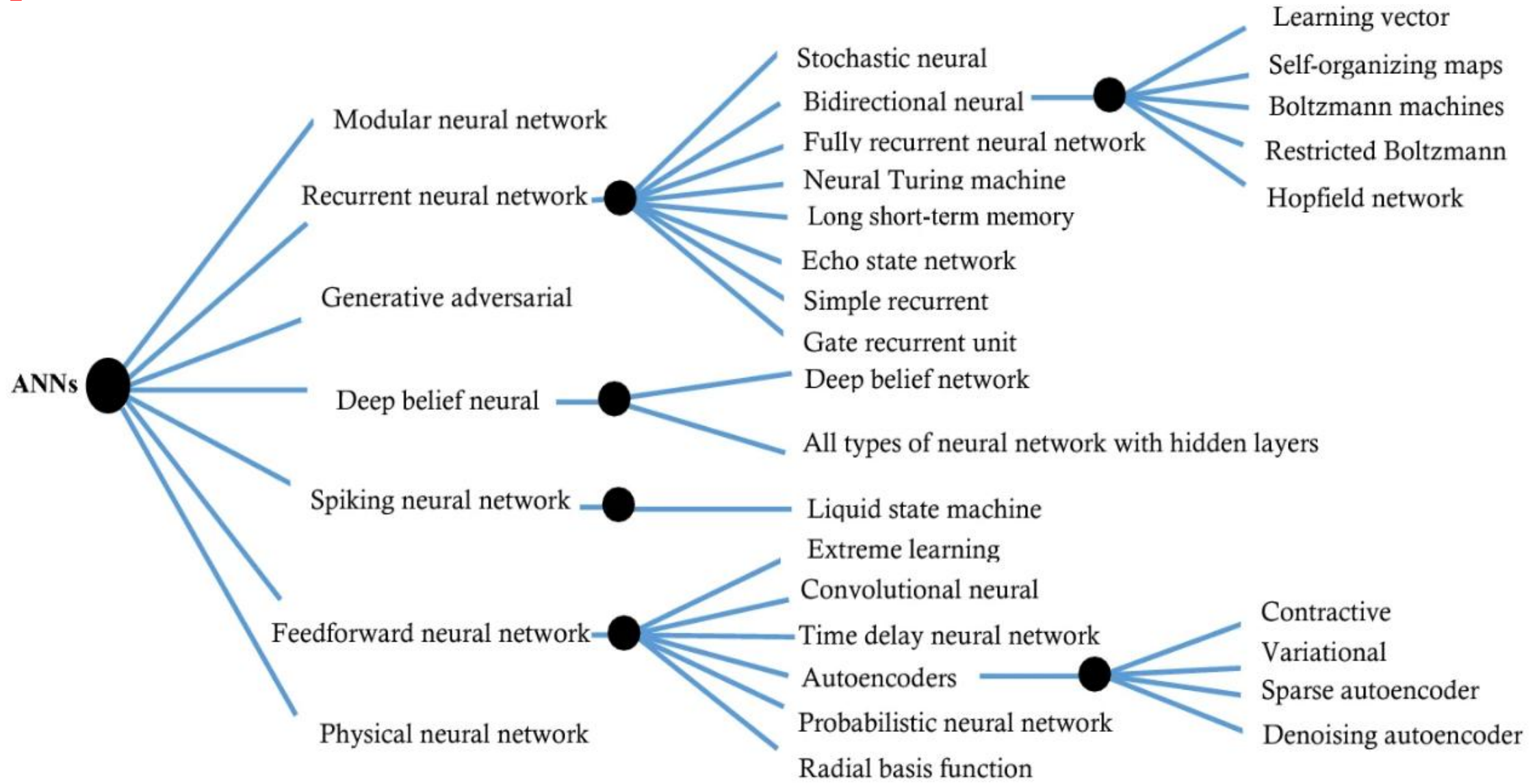
2.**ReLU (Rectified Linear Unit)**: A popular choice for hidden layers, it returns the input if positive and zero otherwise. It helps to solve the [vanishing gradient problem](#).

3.**Tanh (Hyperbolic Tangent)**: Similar to sigmoid but outputs values between -1 and 1. It is used in hidden layers when a broader range of outputs is needed.

4.**Softmax**: Converts raw outputs into probabilities used in the final layer of a network for multi-class classification tasks.

5.**Leaky ReLU**: A variant of ReLU that allows small negative values for inputs helps in preventing “dead neurons” during training.

Types of Artificial Neural Networks



Optimization Algorithms in ANN Training

Optimization algorithms adjust the weights of a neural network during training to minimize errors. The goal is to make the network's predictions more accurate.

Lets see key algorithms:

1.Gradient Descent: Most basic optimization algorithm that updates weights by calculating the gradient of the loss function.

2.Adam (Adaptive Moment Estimation): An efficient version of gradient descent that adapts learning rates for each weight used in deep learning.

3.RMSprop: A variation of gradient descent that adjusts the learning rate based on the average of recent gradients, it is useful in training recurrent neural networks (RNNs).

4.Stochastic Gradient Descent (SGD): Updates weights using one sample at a time helps in making it faster but more noisy.

Applications of Artificial Neural Networks

1.Social Media: ANNs help social media platforms suggest friends and relevant content by analyzing user profiles, interests and interactions. They also assist in targeted advertising which ensures users to see ads tailored to their preferences.

2.Marketing and Sales: E-commerce sites like Amazon use ANNs to recommend products based on browsing history. They also personalize offers, predict customer behavior and segment customers for more effective marketing campaigns.

3.Healthcare: ANNs are used in medical imaging for detecting diseases like cancer and they assist in diagnosing conditions with accuracy similar to doctors. Additionally, they predict health risks and recommend personalized treatment plans.

4.Personal Assistants: Virtual assistants like Siri and Alexa use ANNs to process natural language, understand voice commands and respond accordingly. They help manage tasks like setting reminders helps in making calls and answering queries.

5.Customer Support: ANNs power chatbots and automated customer service systems that analyze customer queries and provide accurate responses helps in improving efficiency in handling customer inquiries.

6.Finance: In the financial industry, they are used for fraud detection, credit scoring and predicting market trends by analyzing large sets of transaction data and spotting anomalies.

Challenges in Deep Learning



Thank you